INFORMATION SYSTEMS QUALITY ASSURANCE

Prepared by: Dr. Hassan Saad

Lecturer at Faculty of Computers and Information



SOFTWARE QUALITY & TESTING



DATA INTEGRITY & ACCURACY



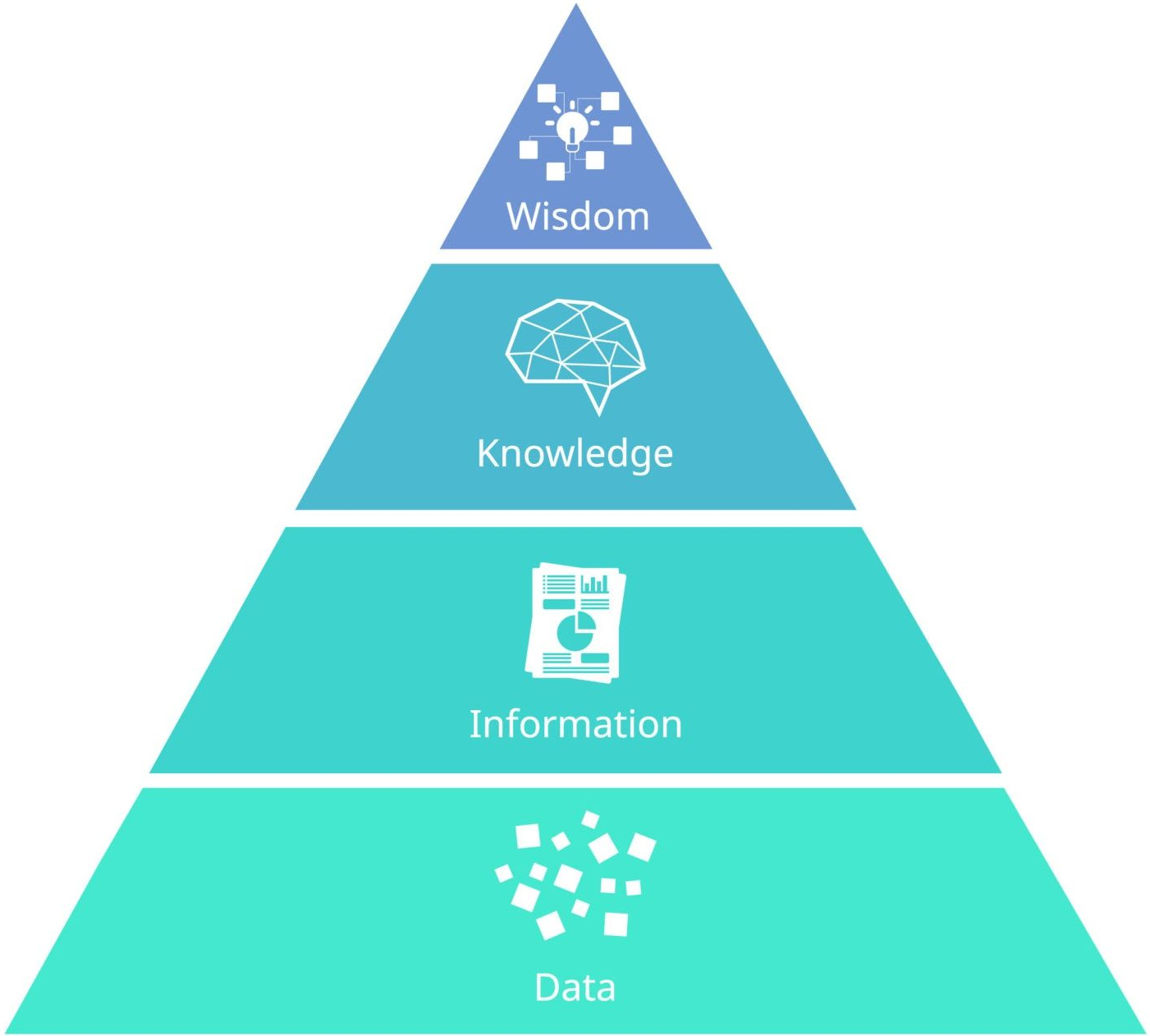
INFORMATION SECURITY



CLOUD ARCHITECTURE & RELIABILITY

DIKW Hierarchy

- **Data**
  - Unorganized, raw symbols, facts, or observations without context.
- **Information**
  - Processed, structured data given context, meaning, and purpose.
- **Knowledge**
  - Synthesized information internalized by humans through experience, learning, and reasoning to enable action.


Wisdom
Knowledge
Information
Data

Data Quality Dimensions

- **Accuracy**

The degree to which data correctly reflects the real-world object or event it represents.

- **Completeness**

The extent to which all required data values are present without missing attributes.

- **Consistency**

The absence of contradictions across different datasets, tables, or systems.

Data Quality Dimensions

- **Timeliness**

How up-to-date the data is relative to when the real-world event occurred.

- **Reliability**

The trust placed in data based on steady, repeatable collection and pipeline stability.

- **Relevance**

The applicability and utility of the data for a specific user, task, or business decision.

Data Quality Dimensions

- **Accessibility**

The ease and speed with which authorized users can retrieve and query the data.

- **Security**

The protection of data against unauthorized access, leaks, corruption, or tampering.

Trade-offs

- Timeliness vs. Accuracy
- Accessibility vs. Security
- Completeness vs. Relevance

Information System

An Information System (IS) is an integrated set of components for collecting, storing, processing, and communicating data and information to support decision-making

Components of IS

- **Hardware:** The physical equipment used to process and store data (e.g., servers, computers, network devices).
- **Software:** The programs and applications that govern processing (e.g., operating systems, DBMS, business applications).

Components of IS

- **Data:** The raw facts and figures stored in databases before being processed into meaningful information.
- **People:** The users, analysts, developers, and administrators who operate and interact with the system.
- **Procedures:** The rules, policies, and workflows that define how data is collected, processed, and utilized.

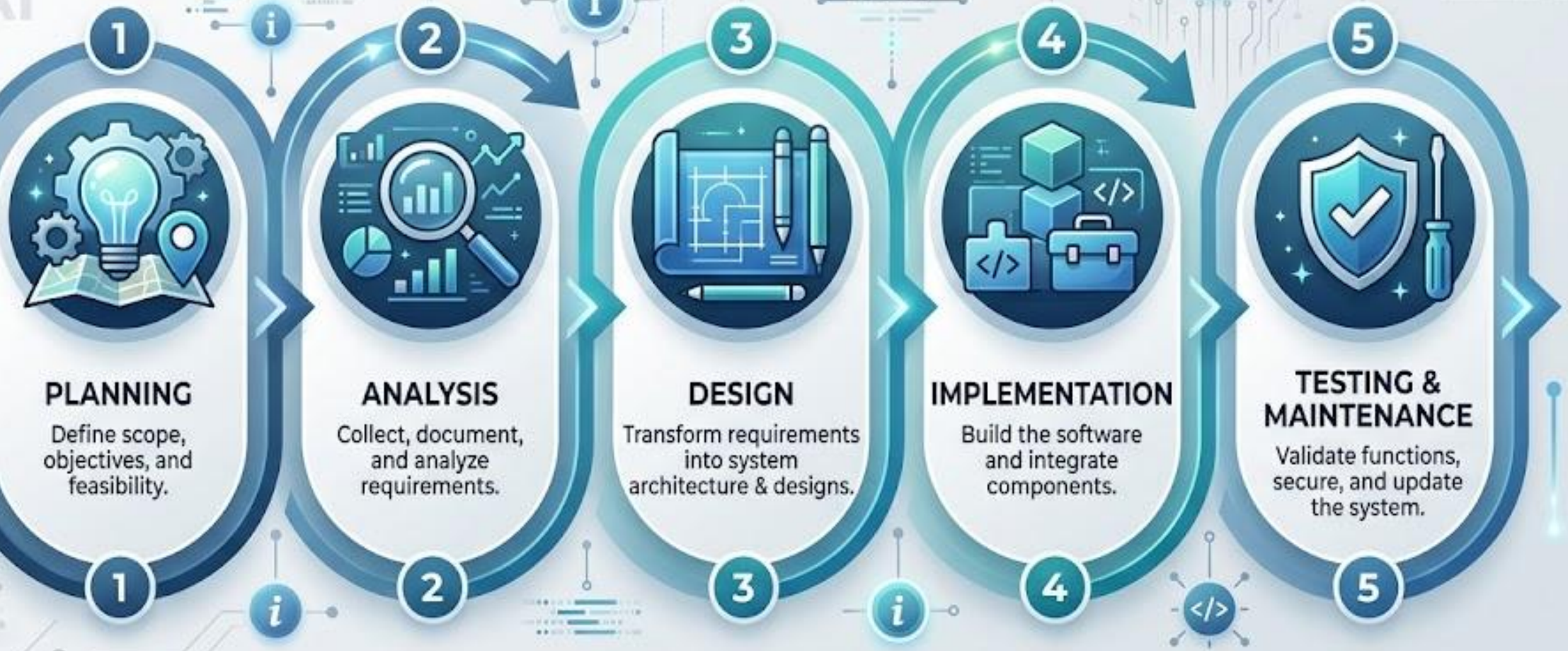
Information System Overview

An Information System (IS) is a unified and integrated set of interconnected components that work together to collect, store, process, and communicate data and information. Its primary purpose within an organization is to support effective decision-making, coordination, control, analysis, and visualization of operations.

Information System Overview

To achieve this, it relies on a specific combination of interrelated elements, including physical hardware, application software, structured data, human resources, and the procedures that govern operations. In essence, the system transforms raw data, gathered from various internal and external sources, into valuable information that is presented as useful output, often incorporating a feedback loop to continuously refine input and improve system accuracy and relevance.

FIVE-STAGE SYSTEM DEVELOPMENT LIFE CYCLE (SDLC)



A structured process for building quality systems.

Quality

Quality is the degree to which a set of inherent characteristics fulfills requirements.

Garvin's 8 Dimensions of Quality

- Performance
- Features
- Reliability
- Conformance
- Durability / Scalability
- Serviceability / Maintainability
- Aesthetics
- Perceived Quality

Information Systems Quality

Information Systems Quality ensures that information systems meet organizational objectives, user needs, and defined requirements by maintaining the quality of their software, data, processes, people, technology, and services.

Information Systems Quality

A high-quality information system provides accurate, complete, consistent, timely, reliable, secure, and relevant information while supporting effective performance and decision-making.

Information Systems Quality

The course integrates quality assurance, quality control, software and information quality, testing, quality measurement, standards, and continuous improvement throughout the system lifecycle.

Information Systems Quality

Information Systems Quality is the degree to which an information system meets defined requirements, user needs, and organizational objectives while providing reliable, accurate, secure, and useful information and services.

Software Quality

Software Quality is the degree to which software meets specified requirements, satisfies user needs, and performs its intended functions effectively, reliably, and securely.

SQ VS ISQ

Software Quality focuses specifically on the software product and its technical requirements, whereas Information Systems Quality covers the broader ecosystem, including software, hardware, data, people, processes, and infrastructure.

SQ VS ISQ

Software quality is a key subset of IS quality, but excellent software alone does not ensure a high-quality information system. A quality IS must also deliver accurate data, streamline business processes, satisfy users, and fulfill organizational goals.